

PRACTICAL FILE

Programming for Problem Solving using C

Experiment

Menu-Driven C Program (using Switch-Case) to find the Sum of Squares, Sum of Cubes, and the Difference between them, for the digits of a positive integer.

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1. Aim

To write, compile, and execute a menu-driven C program using switch-case that inputs a positive integer from the user and, based on the user's choice, computes:

1. The sum of squares of its digits, using a while loop.
2. The sum of cubes of its digits, using a for loop.
3. The difference between the sum of cubes and the sum of squares of its digits, using a do-while loop.

2. Algorithm

Step 1: Start.

Step 2: Display a menu with three choices: 1 -> Sum of squares of digits 2 -> Sum of cubes of digits 3 -> Difference of (sum of cubes - sum of squares)

Step 3: Read the user's choice into an integer variable 'choice'.

Step 4: Read a positive integer into variable 'number'. If the number entered is not positive, display an error message and stop.

Step 5: Use a switch statement on 'choice':

Case 1 (while loop):

- 5.1 Set $n = \text{number}$ and $\text{sum} = 0$.
- 5.2 While n is not 0: extract $\text{digit} = n \% 10$, add $\text{digit} * \text{digit}$ to sum , then set $n = n / 10$.
- 5.3 Print sum as the sum of squares of digits.

Case 2 (for loop):

- 5.4 For $n = \text{number}$; n is not 0; $n = n / 10$: extract $\text{digit} = n \% 10$ and add $\text{digit} * \text{digit} * \text{digit}$ to sum .
- 5.5 Print sum as the sum of cubes of digits.

Case 3 (do-while loop):

- 5.6 Set $n = \text{number}$, $\text{sumSquares} = 0$, $\text{sumCubes} = 0$.
- 5.7 Do: extract $\text{digit} = n \% 10$; add $\text{digit} * \text{digit}$ to sumSquares ; add $\text{digit} * \text{digit} * \text{digit}$ to sumCubes ; set $n = n / 10$; repeat while n is not 0.
- 5.8 Compute $\text{difference} = \text{sumCubes} - \text{sumSquares}$ and print it.

Default: Display an "Invalid choice" message.

Step 6: Stop.

3. Test Case Table

The program was compiled and executed for the following test cases, covering the given examples, boundary conditions (single digit, digits containing zero), and invalid inputs.

TC#	Choice	Input	Description	Expected Output	Actual Output	Result
1	1	123	Sum of squares of digits (while loop)	Sum of squares = 14	Sum of squares of digits = 14	PASS
2	2	123	Sum of cubes of digits (for loop)	Sum of cubes = 36	Sum of cubes of digits = 36	PASS
3	3	123	Difference of cubes & squares (do-while loop)	Difference = 22	Cubes=36, Squares=14, Difference = 22	PASS
4	1	9	Single-digit number, while loop	Sum of squares = 81	Sum of squares of digits = 81	PASS

TC#	Choice	Input	Description	Expected Output	Actual Output	Result
5	3	100	Number containing zero digits	Cubes=1, Squares=1, Diff=0	Cubes=1, Squares=1, Difference = 0	PASS
6	4	55	Invalid menu choice (boundary/negative test)	"Invalid choice" message	Invalid choice! Please select 1, 2, or 3.	PASS
7	1	-5	Invalid input: non-positive number	"Invalid input" message	Invalid input! Please enter a positive integer.	PASS

4. Source Code (digit_ops.c)

The program below was written and saved as digit_ops.c, then compiled and executed as shown in Section 5.

```

/* =====
Program   : digit_ops.c
Purpose   : Menu-driven program using switch-case to:
            1. Find sum of squares of digits (while loop)
            2. Find sum of cubes of digits (for loop)
            3. Find difference of (sum of cubes - sum of squares)
               (do-while loop)

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===== */

#include <stdio.h>

int main()
{
    int choice, number, n, digit;
    long sumSquares, sumCubes, difference;

    printf("=====\n");
    printf("    DIGIT OPERATIONS MENU PROGRAM\n");
    printf("=====\n");
    printf("1. Sum of squares of digits (while loop)\n");
    printf("2. Sum of cubes of digits (for loop)\n");
    printf("3. Difference (cubes - squares) (do-while loop)\n");
    printf("=====\n");
    printf("Enter your choice (1-3): ");
    scanf("%d", &choice);

    printf("Enter a positive integer: ");
    scanf("%d", &number);

    /* Validate positive integer */
    if (number <= 0)
    {
        printf("Invalid input! Please enter a positive integer.\n");
        return 0;
    }

    switch (choice)
    {
        case 1:
            /* Sum of squares of digits using while loop */
            n = number;

```

```

        sumSquares = 0;
        while (n != 0)
        {
            digit = n % 10;
            sumSquares = sumSquares + (digit * digit);
            n = n / 10;
        }
        printf("Number: %d, Sum of squares of digits = %ld\n",
            number, sumSquares);
        break;

    case 2:
        /* Sum of cubes of digits using for loop */
        n = number;
        sumCubes = 0;
        for ( ; n != 0; n = n / 10)
        {
            digit = n % 10;
            sumCubes = sumCubes + (digit * digit * digit);
        }
        printf("Number: %d, Sum of cubes of digits = %ld\n",
            number, sumCubes);
        break;

    case 3:
        /* Difference of sum of cubes and sum of squares using do-while loop */
        n = number;
        sumSquares = 0;
        sumCubes = 0;
        do
        {
            digit = n % 10;
            sumSquares = sumSquares + (digit * digit);
            sumCubes = sumCubes + (digit * digit * digit);
            n = n / 10;
        } while (n != 0);

        difference = sumCubes - sumSquares;
        printf("Number: %d, Sum of Cubes = %ld, Sum of Squares = %ld, "
            "Difference = %ld\n",
            number, sumCubes, sumSquares, difference);
        break;

    default:
        printf("Invalid choice! Please select 1, 2, or 3.\n");
}

return 0;
}

```

5. Compilation and Execution (VS Code Terminal)

The program was compiled and run directly from the integrated terminal in Visual Studio Code using the GCC compiler, as shown below:

```

PS C:\Users\Rudra Pratap Singh> gcc digit_ops.c -o digit_ops

PS C:\Users\Rudra Pratap Singh> .\digit_ops.exe

```

```

=====
      DIGIT OPERATIONS MENU PROGRAM
=====
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference (cubes - squares) (do-while loop)
=====
Enter your choice (1-3): 1
Enter a positive integer: 123
Number: 123, Sum of squares of digits = 14

PS C:\Users\Rudra Pratap Singh> .\digit_ops.exe
Enter your choice (1-3): 2
Enter a positive integer: 123
Number: 123, Sum of cubes of digits = 36

PS C:\Users\Rudra Pratap Singh> .\digit_ops.exe
Enter your choice (1-3): 3
Enter a positive integer: 123
Number: 123, Sum of Cubes = 36, Sum of Squares = 14, Difference = 22

PS C:\Users\Rudra Pratap Singh>

```

6. Additional Test Runs (Edge Cases)

```

PS C:\Users\Rudra Pratap Singh> .\digit_ops.exe
Enter your choice (1-3): 1
Enter a positive integer: 9
Number: 9, Sum of squares of digits = 81

PS C:\Users\Rudra Pratap Singh> .\digit_ops.exe
Enter your choice (1-3): 3
Enter a positive integer: 100
Number: 100, Sum of Cubes = 1, Sum of Squares = 1, Difference = 0

PS C:\Users\Rudra Pratap Singh> .\digit_ops.exe
Enter your choice (1-3): 4
Enter a positive integer: 55
Invalid choice! Please select 1, 2, or 3.

PS C:\Users\Rudra Pratap Singh> .\digit_ops.exe
Enter your choice (1-3): 1
Enter a positive integer: -5
Invalid input! Please enter a positive integer.

PS C:\Users\Rudra Pratap Singh>

```

7. Result

The menu-driven C program was successfully written, compiled without errors or warnings, and executed for all planned test cases. The program correctly computed the sum of squares of digits (while loop), the sum of cubes of digits (for loop), and the difference between the sum of cubes and the sum of squares (do-while loop). All test cases, including boundary and invalid-input cases, produced the expected output. Hence, the program works correctly for all cases.